Response to Reviewers — Second Revision

Manuscript: A Four-Dimension Pre-Modeling Audit Protocol for Educational Prediction Benchmarks
Journal: Scientific Reports
Revision: Second round

Dear Editor and Reviewers,

We thank the reviewers for their careful reading of our revised manuscript. Below we address each remaining comment point by point. All changes are described with specific line references.

Response to Reviewer 1

Recommendation: Accept with minor suggestion.

Comment: The authors may wish to cite Saşmaz (2026), which uses a multi-model machine learning framework on PISA 2022 data and is relevant to the benchmarking methodology discussed here.

Response: We thank the reviewer for this suggestion. We have added a critical citation of **?** to the Discussion section (lines 351–352). The added sentence notes that, while **?** applies a multi-model framework with stratified k-fold cross-validation to PISA 2022 data, it does not report group-aware holdout results, illustrating the persistent gap between current evaluation practice and the audit standards proposed here. We believe this framing strengthens rather than dilutes our argument.

Response to Reviewer 2

Recommendation: Minor revision (second round).

We thank Reviewer 2 for the close reading and for pushing us to clarify four important points.

Comment 2-1: Feature set alignment for fair retention-ratio calculation

Reviewer: The retention ratio currently compares group-holdout R^2 (with group identifiers excluded) against iid R^2 (with group identifiers included). This asymmetry needs to be addressed.

Response: We agree and have made the following changes:

- Modified `run_7dataset_audit.py` (lines 238–244) to run IID splits on the same clean feature matrix (group-identifier columns excluded) when the adapter reports such columns. The clean iid R^2 is stored as `iid_r2_clean` in the output profile.
- Updated the retention-ratio logic (line 337) to use `iid_r2_clean` when available, ensuring apples-to-apples comparison.

- Re-ran the full pipeline and updated all affected numbers:
  - xAPI-Edu: iid R^2 0.599 $\rightarrow$ 0.614, retention 81% $\rightarrow$ 79% (profile unchanged: Strong)
  - UCI Student: iid R^2 0.262 $\rightarrow$ 0.242, retention -37% $\rightarrow$ -40% (profile unchanged: Mostly Passing)
  - Entrance Exam: iid R^2 unchanged at 0.438 (board features have negligible impact on iid performance)
- Updated all affected numbers in the main text (xAPI-Edu 0.614, UCI Student 0.242, retention ratios 79% and -40%).
- Removed stale values (0.599, 0.262, 81%, -37%) from the manuscript.
- Updated `results/audit_7dataset_results.json` with the corrected canonical values.
- Regenerated all figures.

No profile boundaries were crossed for any dataset, and the three key substantive conclusions remain unchanged.

Comment 2-2: Structural versus model-dependent fragility

Reviewer: Ridge and ensemble models recover group-holdout performance for Entrance Exam. Failure of the unregularized linear model alone is insufficient to establish structural fragility. Persistent failures across model families should be distinguished from model-dependent or partly remediable failures.

Response: We fully agree and have revised the manuscript throughout to draw this distinction clearly:

- **Finding 1** (line 333): Now explicitly distinguishes "structural fragility" (Higher Ed: all models $R^2 < -8$; UCI Student: linear -0.097, RF 0.018, GBT -0.057) from "linear-model-only failure that is partly remediable by ensemble models" (Entrance Exam: RF $R^2 \approx 0.32$ vs. linear -0.060).
- **Finding 2** (line 335): Retitled to "Structural fragility persists across model families, while model-dependent failures are partly remediable." Includes Entrance Exam's ensemble recovery as a counterexample.
- **Finding 4** (line 339): Now notes that Entrance Exam's iid-to-group gap is "partly closed by ensemble models."
- **Conclusion (1)** (line 369): Separates structural collapse from model-dependent failure.
- **Conclusion (2)** (line 371): Retitled from "Fragility persists across model families" to "Structural collapse versus model-dependent failure."
- **Detailed-results section** (line 194): Adds sentence: "The nature of the failure differs: UCI Student and Higher Ed exhibit structural fragility (all model families collapse), whereas Entrance Exam's failure is linear-model-specific and partly remediable—its RF group R^2 reaches approximately 0.32."
- **Table 1 caption** (line 188): Replaced "flags the structural fragility of the base benchmark" with language describing the audit's role as flagging the weakest link, with the detailed breakdown distinguishing structurally invariant collapse from model-dependent failure.
- **Line 260**: Replaced "fragile or partially fragile datasets" with "datasets with metadata-adequacy failures" and annotated each dataset's failure pattern.
- Remaining uses of "fragility" in the abstract and general discussion are retained as high-level descriptors of the cross-group generalization failure phenomenon.

Comment 2-3: Feature-attribution scope limitation

Reviewer: The feature-attribution analysis is computed under iid splits only. The connections drawn to audit dimensions are interesting but not directly supported. Either compute group-aware attributions or clearly limit scope.

Response: We have chosen to clearly limit the scope rather than write new group-aware attribution code. At the end of the feature-attribution section (line 322), we added:

"We note that this feature-attribution analysis is conducted under iid random splits and does not directly evaluate attribution stability under group-aware holdout; the connections drawn to the audit profiles describe the alignment between split-level attribution stability and dataset readiness rather than establishing a causal link to cross-group behavior."

Comment 2-4: RF importance type and feature-name mapping

Reviewer: (a) Report the type of feature importance used (impurity-based or permutation-based). (b) Add a table mapping feature codes to meaningful names.

Response: (a) In the feature-attribution section (line 318), we now specify: "Random Forest feature importances (impurity-based, using scikit-learn's default `feature_importances_`)."

(b) We added a new Supplementary Table S7 (page 9 of the supplementary PDF) mapping all feature codes (e.g., f5, f28, f3) used in Tables S5–S6 to human-readable feature names for UCI Student, Higher Ed, and OULAD. The table covers every feature that appears in the top-10 coefficient and importance tables.

Response to Reviewer 3

Recommendation: Not assigned to this revision.

Response: We thank Reviewer 3 for the detailed comments. Before addressing the substance, we provide a brief overview of our manuscript to correct what appears to be a mismatch in the review assignment.

Our manuscript in brief:

- *Title:* Limited Structural Reliability in Public Educational Prediction Benchmarks: A Four-Dimension Audit of Seven Datasets
- *Keywords:* educational prediction, benchmark auditing, group-aware validation, dataset reliability, learning analytics
- *Research object:* Seven public educational prediction datasets (N = 145 to 32,593), spanning K-12, higher education, entrance examination, and online learning.
- *Method:* A four-dimension pre-modeling audit protocol: (1) baseline gap, (2) split instability, (3) null separation, (4) metadata adequacy via group-aware holdout. Models: linear regression, ridge, Random Forest, Gradient Boosting. No SHAP, LIME, ELI5, CatBoost, or class-imbalance treatment is involved.
- *Conclusion:* Three of seven datasets pass all four checks. The dominant failure mode is cross-group fragility. Audit profiles are robust to metric choice.

Reviewer 3's comments—focusing on CatBoost, SHAP/LIME/ELI5, class imbalance (76.66%/23.34% split), Borda Count model selection, and a 560,000+ record Moroccan administrative dataset—describe a manuscript that is entirely different from ours. We were therefore unable to address those specific comments in our previous revision, and we apologise that this was not made sufficiently clear at the time.

Notwithstanding this mismatch, the literature on post-modeling interpretability in learning analytics—including work on counterfactual explanations and the effect of class-balancing techniques on explanation quality (**??**)—is directly relevant to the downstream use of datasets like OULAD. That OULAD supports analyses as demanding as counterfactual explanation benchmarking is consistent with its Strong (4/4) audit profile in our study: datasets that fail group-aware holdout are unlikely to provide the stable distributional foundation needed for reliable post-modeling interpretability work. We have added the following sentence to the Discussion (line 352):

*"That OULAD supports post-modeling interpretability analyses as demanding as counterfactual explanation benchmarking (**??**) is consistent with its Strong (4/4) audit profile: datasets that fail group-aware holdout are unlikely to provide the stable distributional foundation needed for reliable post-modeling interpretability work."*

We hope this clarifies the scope of our manuscript and demonstrates our appreciation for the complementary line of work represented by the studies cited above. We remain willing to address any further comments that pertain directly to our submission. We defer to the editor's judgment on whether the current reviewer panel is appropriate for this manuscript.

We hope these revisions fully address all remaining concerns. We are grateful for the reviewers' constructive feedback, which has substantially improved the clarity and precision of the manuscript.

Respectfully,
Yan Ma Lizhuo Zhang